Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

Department of Artificial Intelligence

Agentic AI (IA462) —Semester VIII

Rules and Regulations

Practice Manual — Building AI Agents from First Principles

Scheme: L–T–P 3–0–0, 3 Credits (2024–25 Curriculum)

1. Objective and Scope

In this manual students build the core mechanisms of agentic AI themselves — the perception–action loop, deliberative planning, tool use and function calling, the ReAct pattern, retrieval-augmented generation, memory and state, multi-agent coordination, and a learning agent — each as a short, complete, runnable program. Working underneath the frameworks makes the behaviour of tools such as LangGraph, AutoGen and CrewAI legible: every pattern those frameworks package appears here as about forty transparent lines of Python.

IA462 carries no scheduled practical hours. This manual is therefore prescribed for tutorial, assignment and self-study use, and the rules below govern all work submitted against it.

2. Offline-First Policy

Every reference program in Experiments 1 to 8 must run fully offline — no API key, no network access and no paid account. Where a large language model sits in the production version of a pattern (function calling, ReAct), the programs use a clearly marked scripted stub occupying exactly the LLM's position in the loop, so that the surrounding machinery — the part students must learn to build — is the real thing. Each such experiment carries one optional exercise that swaps the stub for a real LLM API call; these exercises require an API key and are not required for completion of the manual.

Experiment 9 is the single exception, and is governed by clause 3.3 below.

3. Permitted and Prohibited Tools

3.1 Experiments 1 to 8. Agent frameworks and orchestration libraries are prohibited. Every loop, dispatcher, parser, retriever, memory store and protocol must appear explicitly in the student's own code. A submission that delegates any of these to a framework will not be evaluated for the implementation component, irrespective of whether the program runs correctly. The permitted software is listed in Table 1.

Table 1. Software policy for Experiments 1 to 8.

Status	Tools
Permitted	The Python standard library (json, collections, datetime, random), NumPy, pandas and Matplotlib
Permitted (utility only)	scikit-learn — TfidfVectorizer and cosine_similarity, in Experiment 5 only
Permitted (restricted)	gymnasium — for the final exercise of Experiment 8 only
Prohibited	Agent frameworks and orchestration libraries: LangChain, LangGraph, AutoGen, CrewAI, LlamaIndex and similar

3.2 Experiment 9. LangGraph and CrewAI are permitted, and required, in Experiment 9 only. Access is conditional: the from-scratch versions of Experiment 4 (ReAct) and Experiment 7 (Contract Net Protocol) must already have been submitted and evaluated. The deliverable of Experiment 9 is the comparison — what the framework does that you wrote, what it does differently, and what it conceals — not a working demonstration.

3.3 Offline provision for Experiment 9. Frameworks may be used without a paid API key. Either drive them with a locally served model, or place the scripted stub of Experiment 4 in the reasoning node; LangGraph accepts an arbitrary callable in place of a chat model, so the graph runs unchanged. Students who choose to use a hosted API do so at their own cost and must state so in the record.

4. Conduct of Work

1. Study the theory and algorithm sections first; viva questions assume this preparation.
2. Type and run the reference program and reproduce the sample output before attempting the exercises. Programs using random seeds reproduce exactly; remove the seed to observe behavioural variety.
3. Complete at least two exercises per experiment. LLM-API exercises are optional throughout.
4. Record aim, algorithm, program, output and conclusions for each experiment.

5. Record and Submission

1. Save each file as LAB_No_Roll_No.doc — for example, LAB_1_U21AI001.doc.
2. Paste a snapshot of each step followed along with the result(s); output must be from the student's own execution, not copied from the manual.
3. Record observations and comments after the results, in the student's own words.
4. Submit on or before the notified date. Late submission forfeits the record and timely submission component of the rubric.

6. Assessment

Each experiment carries 10 marks, distributed as follows.

Table 2. Assessment rubric, per experiment.

Component	Marks
Working implementation and correctness of the agent mechanism	4
Exercises attempted, traces and analysis	3
Viva — conceptual understanding	2
Record and timely submission	1
Total	10

For Experiment 9 the implementation component is awarded for the accuracy and depth of the component-by-component mapping, not for the framework code, which is largely boilerplate.

7. Academic Integrity

All code, traces and written analysis must be the student's own work. Discussion of concepts is encouraged; sharing or copying of code, output snapshots or observations is not. Any external source consulted must be cited in the record. Copied submissions are liable to be awarded zero for the experiment and reported under the Institute's academic misconduct procedure.